

HACKATHON OPERATIONS HUB

Database Schema Documentation

PostgreSQL Database Architecture

28

TABLES

10

MODULES

7

JUNCTION TABLES

Version 1.0 - August 3, 2026

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Overview

Database: PostgreSQL
Total Tables: 28
Modules: 10
Architecture: Raw SQL (No ORM)

Purpose: Complete database schema for the Hackathon Operations Hub platform

Module 1: Users & Profiles

users				Table 1
COLUMN	TYPE	CONSTRAINTS	DESCRIPTION	
id	BIGSERIAL	PRIMARY KEY	Unique user identifier	
email	VARCHAR(255)	UNIQUE, NOT NULL	User email address	
password_hash	VARCHAR(255)	NOT NULL	Bcrypt hashed password	
full_name	VARCHAR(150)	NOT NULL	User's full name	
role	VARCHAR(20)	NOT NULL, DEFAULT 'participant'	User role	
is_active	BOOLEAN	NOT NULL, DEFAULT TRUE	Account active status	
created_at	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Account creation timestamp	
CHECK Constraint (role): admin, organizer, participant, judge, volunteer				

profiles				Table 2
Extended user profile information with 1:1 relationship to users.				
COLUMN	TYPE	CONSTRAINTS	DESCRIPTION	
user_id	BIGINT	PRIMARY KEY, FK → users(id) ON DELETE CASCADE	Reference to user	
bio	TEXT	-	User biography	
experience_level	VARCHAR(20)	-	Technical experience level	
github_url	VARCHAR(255)	-	GitHub profile URL	
portfolio_url	VARCHAR(255)	-	Portfolio website URL	
seeking_team	BOOLEAN	NOT NULL, DEFAULT FALSE	Whether user is looking for a team	
updated_at	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Last profile update	
CHECK Constraint (experience_level): beginner, intermediate, advanced				

skills

Table 3

Master list of technical skills available in the system.

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
id	BIGSERIAL	PRIMARY KEY	Unique skill identifier
name	VARCHAR(100)	UNIQUE, NOT NULL	Skill name

user_skills

Table 4 (Junction)

Junction table for many-to-many relationship between users and skills.

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
user_id	BIGINT	PRIMARY KEY, FK → users(id) ON DELETE CASCADE	Reference to user
skill_id	BIGINT	PRIMARY KEY, FK → skills(id) ON DELETE CASCADE	Reference to skill

Module 2: Team Formation

team_requests				Table 5
Requests from users seeking team members for hackathon projects.				
COLUMN	TYPE	CONSTRAINTS	DESCRIPTION	
id	BIGSERIAL	PRIMARY KEY	Unique request identifier	
user_id	BIGINT	UNIQUE, NOT NULL, FK → users(id) ON DELETE CASCADE	Requesting user	
tech_stack	TEXT	NOT NULL	Preferred technology stack	
goals	TEXT	-	Project goals and objectives	
desired_team_size	INT	NOT NULL, DEFAULT 4	Target team size	
status	VARCHAR(20)	NOT NULL, DEFAULT 'open'	Request status	
created_at	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Request creation time	
CHECK Constraint (status): open, matched, cancelled				
CHECK Constraint (desired_team_size): 2-10				

teams				Table 6
Hackathon teams formed for project development.				
COLUMN	TYPE	CONSTRAINTS	DESCRIPTION	
id	BIGSERIAL	PRIMARY KEY	Unique team identifier	
name	VARCHAR(120)	UNIQUE, NOT NULL	Team name	
theme	VARCHAR(150)	-	Project theme or focus area	
capacity	INT	NOT NULL, DEFAULT 4	Maximum team size	
status	VARCHAR(20)	NOT NULL, DEFAULT 'open'	Team status	
created_by	BIGINT	FK → users(id)	Team creator	
created_at	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Team creation time	
CHECK Constraint (status): open, locked, disqualified				
CHECK Constraint (capacity): 1-10				

team_members

Table 7 (Junction)

Junction table for many-to-many relationship between teams and users.

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
team_id	BIGINT	PRIMARY KEY, FK → teams(id) ON DELETE CASCADE	Reference to team
user_id	BIGINT	PRIMARY KEY, FK → users(id) ON DELETE CASCADE	Reference to user
member_role	VARCHAR(50)	NOT NULL, DEFAULT 'member'	Role within team
joined_at	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Timestamp when user joined

Entity Relationship Summary

1:1 One-to-One Relationships

- users ↔ profiles
- teams ↔ projects

1:N One-to-Many Relationships

- users → team_requests
- users → inventory_checkouts (as borrower)
- users → inventory_checkouts (as staff)
- users → damage_reports
- users → certificates
- users → expenses (as approver)
- users → incidents (as reporter)
- users → incidents (as assignee)
- users → incident_comments
- users → feedback_responses
- inventory_items → inventory_checkouts
- inventory_items → damage_reports
- inventory_checkouts → damage_reports
- teams → bookings
- projects → bookings
- projects → project_submissions
- project_submissions → judging_scores
- sponsors → contributions
- venue_areas → bookings
- feedback_forms → feedback_questions

- `feedback_forms` → `feedback_responses`

M:N Many-to-Many Relationships

- `users` ↔ `skills` (via `user_skills`)
- `users` ↔ `teams` (via `team_members`)
- `users` ↔ `sessions` (via `checkins`)
- `users` ↔ `volunteer_shifts` (via `shift_assignments`)
- `feedback_responses` ↔ `feedback_questions` (via `feedback_answers`)

Technical Notes & Best Practices

- **Primary Keys:** All tables use BIGSERIAL for primary keys to support large-scale events
- **Timestamps:** TIMESTAMPTZ used for all timestamp columns to handle timezone conversions
- **Financial Data:** NUMERIC(12,2) used for monetary calculations to avoid floating-point precision issues
- **CASCADE Deletes:** Junction tables use ON DELETE CASCADE to maintain referential integrity
- **Indexes:** Recommended on status fields, date ranges, and foreign keys for query optimization
- **CHECK Constraints:** All enumerated fields use CHECK constraints for data validation
- **Unique Constraints:** Applied to natural keys (`email`, `serial_number`) and composite keys in junction tables

PostgreSQL Database Architecture - 28 Tables across 10 Modules